

STA1023/MAT1063 - Tutorial 4

Question 1

The MGF of Binomial random variable is $M_x(t) = [1 + \theta(e^t - 1)]^n$. Use it to derive the mean and the variance of $X \sim \text{Bin}(n, \theta)$.

Solution:

The moment generating function (MGF) is given by $M_X(t) = [1 + \theta(e^t - 1)]^n$.
The mean of X is the first derivative of the MGF evaluated at $t = 0$:

$$\begin{aligned} M'_X(t) &= \frac{d}{dt} \left([1 + \theta(e^t - 1)]^n \right) \\ &= n[1 + \theta(e^t - 1)]^{n-1} \cdot \frac{d}{dt}(1 + \theta(e^t - 1)) \\ &= n[1 + \theta(e^t - 1)]^{n-1} \cdot \theta e^t \end{aligned}$$

Evaluating at $t = 0$:

$$\begin{aligned} E(X) &= M'_X(0) = n[1 + \theta(e^0 - 1)]^{n-1} \cdot \theta e^0 \\ &= n[1 + \theta(1 - 1)]^{n-1} \cdot \theta(1) \\ &= n(1)^{n-1} \cdot \theta = n\theta \end{aligned}$$

To find the variance, we first need $E(X^2)$, which is the second derivative of the MGF evaluated at $t = 0$. Using the product rule on $M'_X(t) = (n\theta e^t) \cdot [1 + \theta(e^t - 1)]^{n-1}$:

$$\begin{aligned} M''_X(t) &= \frac{d}{dt} \left(n\theta e^t \cdot [1 + \theta(e^t - 1)]^{n-1} \right) \\ &= n\theta e^t \cdot [1 + \theta(e^t - 1)]^{n-1} + n\theta e^t \cdot (n-1)[1 + \theta(e^t - 1)]^{n-2}(\theta e^t) \\ &= n\theta e^t [1 + \theta(e^t - 1)]^{n-1} + n(n-1)\theta^2 e^{2t} [1 + \theta(e^t - 1)]^{n-2} \end{aligned}$$

Evaluating at $t = 0$:

$$\begin{aligned} E(X^2) &= M''_X(0) = n\theta(1)[1 + 0]^{n-1} + n(n-1)\theta^2(1)[1 + 0]^{n-2} \\ &= n\theta + n(n-1)\theta^2 \\ &= n\theta + n^2\theta^2 - n\theta^2 \end{aligned}$$

The variance is given by:

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= (n\theta + n^2\theta^2 - n\theta^2) - (n\theta)^2 \\ &= n\theta - n\theta^2 = n\theta(1 - \theta) \end{aligned}$$

Question 2

If a fair coin is tossed at random five independent times, find the conditional probability of five heads given that there are at least 4 heads.

Solution:

Let X be the number of heads in 5 independent tosses. Since the coin is fair, $X \sim \text{Bin}(n =$

5, $p = 0.5$).

We want to find $P(X = 5 \mid X \geq 4)$. By definition of conditional probability:

$$P(X = 5 \mid X \geq 4) = \frac{P(X = 5 \cap X \geq 4)}{P(X \geq 4)} = \frac{P(X = 5)}{P(X \geq 4)}$$

The probability mass function of a Binomial distribution is $P(X = x) = \binom{n}{x}p^x(1-p)^{n-x}$. Calculate the probabilities:

$$P(X = 5) = \binom{5}{5}(0.5)^5(0.5)^0 = 1 \times (0.5)^5 = \frac{1}{32}$$

$$P(X = 4) = \binom{5}{4}(0.5)^4(0.5)^1 = 5 \times (0.5)^5 = \frac{5}{32}$$

$$P(X \geq 4) = P(X = 4) + P(X = 5) = \frac{5}{32} + \frac{1}{32} = \frac{6}{32}$$

Now, substitute these into the conditional probability formula:

$$P(X = 5 \mid X \geq 4) = \frac{1/32}{6/32} = \frac{1}{6} \approx 0.1667$$

Question 3

A fair coin is tossed 7 times. What is the probability that 6 will appear five or more times?

Solution:

Let X be the number of times a “6” appears in 7 independent rolls of a fair die. Then $X \sim \text{Bin}(n = 7, p = 1/6)$.

We want to find $P(X \geq 5)$, which is the probability of getting 5, 6, or 7 sixes:

$$\begin{aligned} P(X \geq 5) &= P(X = 5) + P(X = 6) + P(X = 7) \\ &= \binom{7}{5} \left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)^2 + \binom{7}{6} \left(\frac{1}{6}\right)^6 \left(\frac{5}{6}\right)^1 + \binom{7}{7} \left(\frac{1}{6}\right)^7 \left(\frac{5}{6}\right)^0 \\ &= 21 \left(\frac{1}{7776}\right) \left(\frac{25}{36}\right) + 7 \left(\frac{1}{46656}\right) \left(\frac{5}{6}\right) + 1 \left(\frac{1}{279936}\right) (1) \\ &= \frac{21 \times 25}{279936} + \frac{7 \times 5}{279936} + \frac{1}{279936} \\ &= \frac{525 + 35 + 1}{279936} = \frac{561}{279936} \approx 0.0020 \end{aligned}$$

Question 4

During a particular period, a university’s information technology office received 20 service orders for problems with printers of which 8 were laser printers and 12 were inkjet models. A sample of 5 of these service orders is to be selected for inclusion in a customer satisfaction survey. If 5 service orders are selected randomly, what is the probability that exactly two of them were inject printers?

Solution:

This is a sampling without replacement problem, so we use the Hypergeometric distribution.

Let $N = 20$ (total number of service orders).

Let $K = 12$ (total number of inkjet orders).

Let $n = 5$ (sample size).

Let X be the number of inkjet printers (spelled “inject” in the prompt) in the sample. We want to find $P(X = 2)$.

The Hypergeometric probability formula is:

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

Substitute the values:

$$P(X = 2) = \frac{\binom{12}{2} \binom{20-12}{5-2}}{\binom{20}{5}} = \frac{\binom{12}{2} \binom{8}{3}}{\binom{20}{5}}$$

Calculate the combinations:

$$\begin{aligned}\binom{12}{2} &= \frac{12 \times 11}{2 \times 1} = 66 \\ \binom{8}{3} &= \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56 \\ \binom{20}{5} &= \frac{20 \times 19 \times 18 \times 17 \times 16}{5 \times 4 \times 3 \times 2 \times 1} = 15504\end{aligned}$$

Substitute these back into the probability equation:

$$P(X = 2) = \frac{66 \times 56}{15504} = \frac{3696}{15504} = \frac{77}{323} \approx 0.2384$$

Question 5

For a basketball player, the probability of making a free throw is 0.7 during a season.

- What is the probability that he makes 3 free throws out of 8 shots?
- What is the probability that he makes his 3rd free throw on his 5th shot?
- What is the probability that he makes his 1st free throw on his 4th shot?

Solution:

Let $p = 0.7$ be the probability of success (making a free throw) and $1 - p = 0.3$ be the probability of failure.

a) Let X be the number of successful free throws in 8 independent shots. $X \sim \text{Bin}(n = 8, p = 0.7)$. We want $P(X = 3)$:

$$\begin{aligned}P(X = 3) &= \binom{8}{3} (0.7)^3 (0.3)^{8-3} \\ &= 56 \times (0.343) \times (0.00243) \\ &= 0.04667544 \approx 0.0467\end{aligned}$$

b) Let Y be the number of shots required to make the 3rd free throw. This follows a Negative Binomial distribution where we want the k^{th} success on the n^{th} trial ($k = 3, n = 5$).

$$\begin{aligned}P(Y = n) &= \binom{n-1}{k-1} p^k (1-p)^{n-k} \\ P(Y = 5) &= \binom{5-1}{3-1} (0.7)^3 (0.3)^{5-3} = \binom{4}{2} (0.7)^3 (0.3)^2 \\ &= 6 \times 0.343 \times 0.09 = 0.18522\end{aligned}$$

c) Let Z be the number of shots required to make the 1st free throw. This follows a Geometric distribution where we want the 1st success on the n^{th} trial ($n = 4$).

$$\begin{aligned} P(Z = n) &= (1 - p)^{n-1}p \\ P(Z = 4) &= (0.3)^{4-1}(0.7) = (0.3)^3(0.7) \\ &= 0.027 \times 0.7 = 0.0189 \end{aligned}$$

Question 6

Let X be the number of typos on a printed page with a mean of 3 typos per page. Find
a) $P(X > 4)$ b) $P(X = 4)$ c) Probability that three randomly selected pages would have more than 8 typos.

Solution:

The random variable X follows a Poisson distribution with mean $\lambda = 3$. The PMF is $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$.

a) $P(X > 4) = 1 - P(X \leq 4) = 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)]$.

$$\begin{aligned} P(X = 0) &= \frac{e^{-3}3^0}{0!} \approx 0.049787 \\ P(X = 1) &= \frac{e^{-3}3^1}{1!} \approx 0.149361 \\ P(X = 2) &= \frac{e^{-3}3^2}{2!} \approx 0.224042 \\ P(X = 3) &= \frac{e^{-3}3^3}{3!} \approx 0.224042 \\ P(X = 4) &= \frac{e^{-3}3^4}{4!} \approx 0.168031 \end{aligned}$$

Summing these gives $P(X \leq 4) \approx 0.815263$.

$$P(X > 4) = 1 - 0.815263 = 0.184737 \approx 0.1847$$

b) As calculated above:

$$P(X = 4) = \frac{e^{-3}3^4}{4!} \approx 0.168031 \approx 0.1680$$

c) Let Y be the number of typos on 3 randomly selected pages. Since the Poisson distribution is additive, the new mean is $\lambda_{new} = 3 \times \lambda = 3 \times 3 = 9$. So, $Y \sim Poisson(9)$.

We want to find $P(Y > 8) = 1 - P(Y \leq 8) = 1 - \sum_{y=0}^8 \frac{e^{-9}9^y}{y!}$.

$$\begin{aligned} P(Y \leq 8) &\approx 0.00012 + 0.00111 + 0.00500 + 0.01499 + 0.03374 + 0.06073 + 0.09109 + 0.11712 + 0.13176 \\ &\approx 0.455653 \end{aligned}$$

$$P(Y > 8) = 1 - 0.455653 = 0.544347 \approx 0.5443$$

Question 7

An experiment of drawing a random card from an ordinary playing cards deck is done with replacing it back. This was done ten times. Find the probability of getting 2 spades, 3 diamond, 3 clubs and 2 hearts.

Solution:

This is a Multinomial distribution problem. A standard deck has 52 cards, with 13 cards in each of the 4 suits (Spades, Diamonds, Clubs, Hearts).

Since the draws are with replacement, the probability of drawing a specific suit on any given draw is constant:

$$p_{spades} = p_{diamonds} = p_{clubs} = p_{hearts} = \frac{13}{52} = \frac{1}{4}$$

Let $n = 10$ be the total number of draws. Let $x_1 = 2, x_2 = 3, x_3 = 3, x_4 = 2$ be the number of spades, diamonds, clubs, and hearts respectively.

The probability formula for the Multinomial distribution is:

$$P(X_1 = x_1, \dots, X_k = x_k) = \frac{n!}{x_1!x_2!\dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

Substituting our values:

$$\begin{aligned} P(2S, 3D, 3C, 2H) &= \frac{10!}{2!3!3!2!} \left(\frac{1}{4}\right)^2 \left(\frac{1}{4}\right)^3 \left(\frac{1}{4}\right)^3 \left(\frac{1}{4}\right)^2 \\ &= \frac{3628800}{(2)(6)(6)(2)} \left(\frac{1}{4}\right)^{10} \\ &= \frac{3628800}{144} \left(\frac{1}{1048576}\right) \\ &= 25200 \times \frac{1}{1048576} = \frac{25200}{1048576} \approx 0.02403 \end{aligned}$$

Question 8

During a certain month, a company sold 10,000 new watches. Past experience indicates that the probability that a new watch will need repair during its warranty period is 0.002. Compute (Using Poisson Approximation), the probability that:

- zero watches will need warranty work.
- no more than 5 watches will need warranty work.
- no more than 10 watches will need warranty work.
- no more than 20 watches will need warranty work.

Solution:

Let X be the number of watches needing warranty work. X inherently follows a Binomial distribution with $n = 10,000$ and $p = 0.002$. Since n is very large and p is very small, we use the Poisson approximation with parameter λ :

$$\lambda = np = 10000 \times 0.002 = 20$$

Therefore, $X \approx \text{Poisson}(\lambda = 20)$. The PMF is $P(X = x) = \frac{e^{-20} 20^x}{x!}$.

a) Zero watches:

$$P(X = 0) = \frac{e^{-20} 20^0}{0!} = e^{-20} \approx 2.061 \times 10^{-9}$$

b) No more than 5 watches:

$$\begin{aligned} P(X \leq 5) &= \sum_{x=0}^5 \frac{e^{-20} 20^x}{x!} = e^{-20} \left(1 + 20 + \frac{20^2}{2} + \frac{20^3}{6} + \frac{20^4}{24} + \frac{20^5}{120} \right) \\ &\approx 7.16 \times 10^{-5} \text{ (or } 0.0000716) \end{aligned}$$

c) No more than 10 watches:

$$P(X \leq 10) = \sum_{x=0}^{10} \frac{e^{-20} 20^x}{x!} \approx 0.01081$$

d) No more than 20 watches:

$$P(X \leq 20) = \sum_{x=0}^{20} \frac{e^{-20} 20^x}{x!} \approx 0.5591$$